

Optical Integrated Sensing and Communication for 6G Through a Systematic Review of Architectures, Metrics, and Tradeoffs Across Optical Platforms

Fatih Dönmez, Ahmet Altuncu, and Mustafa Namdar

Abstract—Light carries data and signatures of the path and objects it encounters, allowing the same optical infrastructure to support communication and sensing. Optical integrated sensing and communication (O-ISAC) builds on this dual capability across fiber, free space optical, visible light, and light fidelity (LiFi) platforms. Photonic hardware extends the concept to millimeter wave and terahertz systems, while hybrid designs connect physical domains. This breadth motivates comparison of the design insights that extend across platforms and those tied to a particular physical setting. A systematic literature search covering 1 January 2020 through 22 June 2026 identified 227 eligible reports representing 206 unique studies. We organized this evidence by physical platform and coupling mechanism using structured narrative synthesis and a technical assessment developed for this review. The comparison framework connects each result to its task, measurement plane, operating conditions, and validation setting. Across the included corpus, bandwidth, optical power, hardware constraints, and signal design jointly shape communication reliability and sensing performance. These relationships depend on reported conditions and extend beyond a single rate and resolution frontier. Laboratory studies span diverse architectures. Broader field validation, shared benchmarks, and reusable data and code emerge as priorities for stronger comparison and reconstruction. The resulting taxonomy and framework support the design and evaluation of O-ISAC systems for sixth generation (6G) networks.

Index Terms—6G, integrated sensing and communication, metric comparability, optical communication, optical sensing.

I. INTRODUCTION

The practical appeal of O-ISAC lies in reuse. Optical networks already deploy sources, propagation paths, receivers, and signal processors, and these resources can reveal information about the environment while carrying data. Distributed sensing over fiber, positioning based on visible light, and turbulence monitoring illustrate the breadth of this opportunity [1], [2], [3].

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Sharing turns that opportunity into a systems question. A resource or receiver chosen for one function can change the information available to the other. The central engineering problem is what can be shared and how that sharing shapes system design.

This question has gained importance since integrated sensing and communication was identified as one of the usage scenarios for International Mobile Telecommunications for 2030 and beyond (IMT-2030) [4]. Optical implementations span guided fiber, free space optical and visible light communication links, systems in the millimeter wave and terahertz bands enabled by photonics, and hybrid designs [5], [6], [7]. Their propagation paths, observables, hardware, and performance measures differ. O-ISAC therefore forms a family of related architectures, each with its own physical and measurement context.

That diversity requires a clear survey boundary and careful comparison. The survey includes optical or photonic systems that consider, integrate, design, optimize, or evaluate communication and sensing together. Their common setting may lie in an architecture, optical link, signal or resource framework, hardware platform, channel model, or application scenario. Coupling can appear through a shared waveform, concurrent operation, or another common system choice. Each result is interpreted through its metric definition, measurement plane, and operating conditions. This context supports consistent interpretation and bounded comparison across platforms.

A. Prior Syntheses and Survey Positioning

Prior surveys organize complementary parts of O-ISAC. Each serves a different reader task and uses a different evidence unit and comparison boundary. Table I provides a compact map of these perspectives.

Across these perspectives, synthesis across platforms depends on retaining the physical, measurement, and validation setting of each primary result. This survey addresses that task through 206 unique O-ISAC studies first made publicly available from 1 January 2020 through 22 June 2026.

The survey makes four contributions.

- A taxonomy separates physical modality from the mechanisms that couple communication and sensing.
- A comparison framework connects each performance result to its measurement definition and operating conditions.

TABLE I
REPRESENTATIVE PRIOR O-ISAC SYNTHESIS ORGANIZED BY READER TASK, EVIDENCE UNIT, AND COMPARISON BOUNDARY

Synthesis family	Sources	Optical scope	Evidence unit and comparison logic	Reader task	Boundary
Architecture and system design	[8], [9], [10], [11]	General O-ISAC, free space optical links, retroreflective links, and fiber networks	Architecture, optimization, target, or link model, with taxonomy and relations within each scenario	How are communication and sensing coupled?	Relations remain tied to each architecture and its assumptions.
Analytical limits and estimation methods	[12], [13]	General optical systems and polarization sensing in coherent fiber links	Model, bound, objective, estimator, or learning method, with comparison within a model and among methods	What limits follow from the assumptions?	Empirical interpretation remains tied to the model, modality, and evidence setting.
Technological evolution, standardization, and deployment	[14], [15], [16], [17], [18]	Evolution from radio to optical systems across terrestrial and submarine networks	Historical phase, standardization path, network architecture, or deployment, synthesized by period and ecosystem	How has O-ISAC moved toward deployment?	Infrastructure comparison requires aligned physical and operational conditions.
Physical media and optical networks	[19], [20], [21], [22], [23], [24]	Visible light integrated sensing and communication, guided fiber systems, and optical networks	Medium, observable, sensing path, or network configuration, with taxonomy within related solution classes	How does each path support sensing?	Transfer requires aligned media, observables, paths, and conditions.
Waveforms, resource multiplexing, and photonic hardware	[25], [26], [27], [28], [29], [30]	Fiber, microwave photonics, and millimeter wave or terahertz systems enabled by photonics	Waveform, multiplexing, front end, or processing chain, with a mechanism taxonomy and selected demonstrations	Which mechanisms enable joint operation?	Comparison across photonic systems follows alignment of paths, observables, and metrics.
Applications and challenges across platforms	[31]	Visible light communication, free space optical, fiber, photonics, and hybrid settings	Application, use case, solution class, or challenge, with thematic synthesis across platforms	Which applications and gaps recur?	Interpretation across platforms follows the measurement and operating context of each result.

- A synthesis of recurring tradeoff mechanisms retains the conditions that shape communication and sensing outcomes.
- An assessment of validation, benchmark readiness, and reproducibility grounds research priorities in the reviewed evidence.

The organization of the article follows the analytical sequence of the survey. Sections II and III establish the comparison framework across optical platforms and describe the review methodology and evidence base. Sections IV through VII present the technical synthesis of optical platforms, integration architectures, performance metrics, joint design tradeoffs, validation, enabling technologies, and applications. Section VIII discusses the research priorities and limitations that emerge from this synthesis, and Section IX concludes the article.

II. JOINT-EVIDENCE MODEL AND COMPARISON FRAMEWORK

The value of a cross-platform O-ISAC review does not come from placing studies under platform labels alone. Its central analytical task is to determine when a reported communication result and a sensing result describe a joint engineering relation, which conditions bound that relation, and which conclusion

the evidence permits. We therefore use a relation-centered unit of synthesis. This section defines that unit, specifies the measurement and evidence conditions carried with it, and converts those conditions into explicit comparison permissions. Section III then explains how the framework is populated from the review corpus.

A. What Counts as a Joint-Performance Relation?

Two performance values do not become joint evidence merely because they appear in the same paper or originate from an integrated architecture. The minimum native relation requires a shared or varied design variable, constraint, or impairment to be linked to both the communication and sensing outcomes under an adequate measurement and evidence contract. A complete record that can be tested for transfer additionally resolves the physical system and the coupling mechanism. For relation record r , we express this full unit as

$$J_r = \langle P_r, G_r, X_r, C_r, S_r, M_r, E_r \rangle.$$

Here, P_r identifies the physical context and system boundary; G_r locates what the communication and sensing functions share and where they share it; X_r is the shared or varied design variable, constraint, or impairment linked to both functions; C_r

and S_r are the reported communication and sensing outcomes; and M_r and E_r preserve the measurement contract and evidence envelope. The tuple is not a score, and the presence of a native link does not by itself establish transferability. It is a compact test of whether a claimed relation can be interpreted and assessed without changing the meaning of either result.

The physical context is set by the path that reaches the target, not by the presence of optical hardware alone. The observable modality is therefore kept separate from the technology used to generate, transport, or process the signal. A fiber may itself be the sensing medium or may transport a signal to a downstream target-facing segment. In a photonic-terahertz system, optical components may generate or process the carrier while the target interaction occurs over a radio-frequency path. The corresponding propagation, conversion, illumination, and geometry boundaries are retained for all platform families before comparison in Section IV [2], [7], [32], [33], [34].

The coupling term G_r records where the functions meet; several coupling loci may coexist. The factor X_r is kept separate from that shared object: it is the variable, constraint, or impairment whose setting or exposure is related to both outcomes. This distinction separates *structural integration* from *evidential jointness*. Shared hardware or a common service establishes an architecture relation, whereas an adequate X_r – C_r – S_r trace establishes the minimum native performance relation. Resolving P_r and G_r is additionally required before the complete record can be tested for transfer beyond its native context. Figure 1 shows this anatomy.

A native link is most directly established within one study because its two outcomes can be followed through the same model, experiment, dataset, or operating sweep. Transfer across studies is a separate operation. It is allowed only after the meanings and conditions attached to the native relations have been aligned; co-occurrence at the topic or platform level is not sufficient.

B. Measurement and Evidence Conditions

The link in J_r remains interpretable only while the measurement and evidence contracts remain attached to it. We organize them as

$$M_r = \langle M_C, M_S, A_M \rangle, \quad E_r = \langle E_C, E_S, T_r, B_r \rangle.$$

M_C and M_S retain the definitions used for the communication and sensing quantities, respectively. Each includes the measurement plane, unit, statistic or accounting state, task and target, operating point, channel, geometry, processing conditions, baseline, and ground truth. A_M records whether the condition sets are the same, alignable, materially different, or not yet resolved for comparison. This alignment term prevents a familiar metric name from substituting for an actual measurement match.

Several distinctions are particularly consequential in optical systems. OSNR and electrical SNR refer to different observation planes; launch and received power refer to different positions in the signal chain. Gross and net rate, or error rates before and after correction, use different accounting states. For

sensing, resolution, estimation error, precision, detection probability, and an analytical bound answer different questions. A numerical comparison is therefore formed from matched definitions and references, not from matched abbreviations.

E_C and E_S identify how each outcome was produced, the validation setting, and the exact source locator at which it can be verified. Analytical, simulation, laboratory, offline-dataset, and field evidence are not treated as interchangeable; each supports inference within its own validation setting. T_r preserves the configuration or timing trace that connects the outcomes to X_r , such as a shared sweep, timestamped observation, configuration identifier, or optimization run. B_r states the permitted inference boundary: the native setting or aligned transfer domain within which the relation may be used. A validation label by itself does not create a paired relation: the link must still be recoverable through T_r .

These conditions are deliberately separated from a study-level validation label. A nontransferable result may still be decisive for its source system, and field validation does not repair a mismatch in metric meaning. The framework instead asks a narrower question: what is the strongest conclusion that follows while the original system, measurement, and evidence meanings are preserved? Figure 2 summarizes that decision path.

C. From a Relation Record to a Permitted Claim

The framework converts evidence description into an explicit synthesis decision. The weakest material condition bounds the claim; a strong value on one metric cannot compensate for a missing link or an incompatible definition elsewhere in the relation. Table II states the resulting permission matrix. The matrix separates source-native performance, within-study relations, and cross-study transfer so that the reader can see both what the evidence supports and where that support ends.

When either the independence test or the alignment test for numerical transfer is not satisfied, the decision is explicit non-pooling. The most informative supported non-cross-study use is retained, but the review does not produce a pooled effect, forced unit normalization, platform ranking, or leaderboard. This is a comparison decision, not a judgment that the underlying studies are uninformative. The framework likewise does not manufacture a positive relation merely to fill a synthesis category.

This permission structure supplies the handoff to the remainder of the review. Section III constructs and traces the records; Section IV explains their physical and architectural mechanisms; Section V evaluates the admitted performance relations; Section VI assesses their reconstructability and benchmark readiness; Section VII translates supported mechanisms to application and 6G requirements; and Section VIII converts recurring comparison limits into benchmark and research priorities. The result is a framework that tells an O-ISAC researcher not only where a study belongs, but also which findings can be compared, why that comparison is defensible, and what evidence would enable the next stronger conclusion.

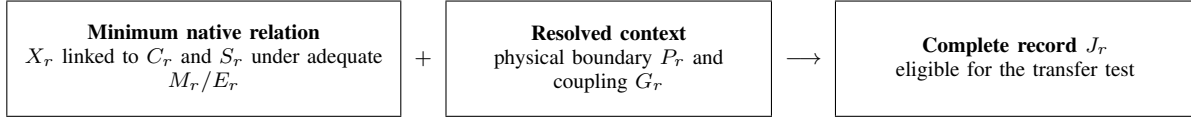


Fig. 1. Native-relation and complete-record requirements. The minimum native relation links a shared or varied factor to both outcomes under an adequate measurement contract and evidence envelope. Resolving the physical and coupling context completes J_r and permits assessment of transfer beyond the source setting; transfer permission still depends on adequate, aligned M/E conditions and independent evidence.

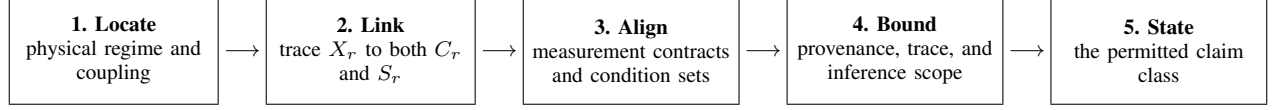


Fig. 2. Claim-permission path used in the review. Each stage preserves the meaning established by the previous stage. An unmet gate does not erase the source result; it routes the record to the strongest permitted class in Table II.

TABLE II
EVIDENCE CONDITIONS AND PERMITTED CLAIM CLASSES IN THE O-ISAC SYNTHESIS

Evidence condition	Permitted claim	Claim boundary	Downstream use
An integration mechanism or shared element is documented, but no traceable $X_r-C_r-S_r$ link is established.	Architecture or feasibility statement.	Shows how the functions can coexist or share resources; it does not establish a joint-performance relation.	Architecture synthesis in Section IV.
A communication or sensing result is source-traceable and its native conditions are recoverable.	Source-native performance statement.	Applies to the reported task, system, and operating region; no effect on the other function is inferred.	Metric landscape in Section V.
One study links X_r to both outcomes through a common trace, A_M confirms the same or alignable condition set, and the M/E contract is adequate.	Within-study relation or tradeoff.	The source-supported direction and any source-reported magnitude apply only to the tested model, experiment, dataset, and operating region.	Native tradeoff synthesis in Section V.
At least two independent studies expose compatible native relations and align physical regime, metric definitions, planes, units, tasks, targets, operating conditions, baselines, validation setting, and relevant uncertainty; remaining differences are bounded.	Bounded cross-study numerical comparison.	Transfer is limited to the shared comparison domain; it is not a universal effect, unconditional pooled estimate, platform ranking, or leaderboard.	Cross-platform synthesis in Section V.
A recurrent, source-supported coupling interpretation connects both functions, but the numerical contracts do not support magnitude transfer.	Mechanism-based or directional statement.	States a source-supported direction or recurring coupling interpretation without assigning a shared numerical effect or unsupported causality.	Mechanism synthesis across Sections IV and V.
A recurring, material omission prevents a decision-relevant relation from being tested or transferred.	Actionable literature-gap statement.	The gap names the missing measurement, baseline, trace, or validation link and the comparison it is designed to test or enable.	Benchmark priorities in Section VI and the roadmap in Section VIII.
No supported relation route is available, or the available routes remain in material conflict.	Audit retention only.	The record remains in the evidence base for transparency and future updating; no reader-facing relational conclusion is drawn from it.	Corpus accounting in Section III.

III. REVIEW PROCESS AND EVIDENCE BASE

With these comparison conditions established, the analysis turns to the evidence base for the technical synthesis. We assembled the corpus through a systematic review reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement. Search reporting followed the PRISMA extension for searches (PRISMA-S) [35], [36]. The following subsections describe corpus construction, evidence extraction, and technical appraisal. Table III maps the analytical role of each evidence category.

A. Corpus Construction

To assemble that evidence base, we searched Scopus, IEEE Xplore, ScienceDirect, SpringerLink, Wiley Online Library, and Taylor & Francis Online. The search covered journal articles and full conference papers that had undergone peer review and were first made publicly available from 1 January 2020 through 22 June 2026. Eligible reports contained sufficient technical content in English, examined communication and sensing within an optical or photonic system, and provided assessable evidence for at least one survey domain. Reviews and other contextual sources supported field positioning. Eligible primary reports proceeded to study reconciliation for

TABLE III

USE OF EXTRACTED EVIDENCE IN THE SURVEY. ROW COUNTS REPRESENT CODING RECORDS, AND STUDY COUNTS MAY OVERLAP ACROSS CATEGORIES.

Evidence use	Rows	Studies	Role in the survey
Quantitative evidence	4,997	206	Numerical or directional synthesis under recorded conditions.
Qualitative evidence	3,206	206	Mechanisms, trends, and limitations.
Claims used for context	31	21	Contextual interpretation reported separately from the primary synthesis.
Claims with source conflicts	72	31	Recorded separately under predefined decision rules, while supported evidence from the same studies remained in the primary synthesis.
Primary synthesis	8,203	206	Quantitative and qualitative evidence combined.

Note. The complete evidence register contains 8,306 rows. Electronic Supplement S-Evidence reports the 8,203 primary synthesis rows and their source locations. Electronic Supplement S-Protocol documents the decision rules.

the technical denominator. Search strategies, eligibility decisions, and protocol documentation are provided in Electronic Supplements S-Search, S-Exclusions, and S-Protocol.

After deduplication, screening, and assessment of full texts, 227 reports met the eligibility criteria. Among them, 21 were companion reports for studies also represented by another eligible report. Consolidating eligible reports by study produced 206 unique studies for the technical synthesis. The totals differ by 21 because companion reports map to studies already represented in the eligible set. This mapping preserves provenance for every report while counting each underlying study once. Electronic Supplement ST-01 cites all 206 studies and traces their 227 reports.

Figure 3 reconciles the selection flow while keeping records, reports, and studies as distinct counting units.

The counting unit changes from records to reports at retrieval and from reports to studies at synthesis. Contextual records support field positioning, while the study set defines the technical denominator.

B. Evidence Extraction and Use

With the study denominator established, we extracted evidence at three linked levels covering the study, report, and individual claim. Each result retained its definition, unit, measurement plane, operating conditions, validation setting, and exact source location. Records remained separate when conditions differed. Each field contained a reported value or a defined availability state. Quantitative extraction used values stated by the source, while graphical findings remained in their reported form. Electronic Supplement S-Data Dictionary provides the complete coding schema.

Claims entered the analysis according to their role. Quantitative and qualitative claims supported the primary synthesis within their recorded contexts. Contextual claims and claims with source conflicts were recorded separately. This structure preserved supported evidence from all 206 studies, while comparisons across studies followed the alignment rules in Table II.

C. Technical Appraisal and Synthesis

To clarify how each study supports the synthesis, we assessed the technical evidence and reporting of all 206 studies with the Technical Quality Assessment Framework (TQAF) developed for this review. The framework applies deterministic rules across eight dimensions. A separate overall contribution

category summarizes the support each study provides to this synthesis. The profile indicates how evidence can support interpretation in its source setting, comparison through a shared benchmark, and comparison across platforms. Electronic Supplements S-Protocol and S-Appraisal provide the framework's methodological scope, scoring rules, and records for each study.

Figure 4 places the eight appraisal dimensions beside the separate overall evidence contribution category. The profile distinguishes support for source interpretation from support for transfer and reuse.

Reporting completeness and metric clarity support interpretation within the source setting. Validation, reproducibility, benchmark readiness, and comparison admissibility introduce further conditions for transfer across studies. These distinct roles explain the profile and guide the structured synthesis.

The structured narrative synthesis retained each result's native task and unit together with its operating conditions and validation setting. This structure preserved the meaning of heterogeneous results while allowing recurring mechanisms and design relationships to be compared across the corpus. The evidence was organized into 115 bodies across seven survey domains, connecting synthesis conclusions to their supporting studies. Electronic Supplement S-Bodies provides their criteria, ratings, and memberships. With this evidence structure in place, the technical synthesis begins with the physical role of optics and the mechanism that couples communication and sensing.

IV. OPTICAL PLATFORMS AND INTEGRATION ARCHITECTURES

In O-ISAC, the physical role of optics determines both where information propagates and where the environment is observed. Optics may carry the user link, illuminate a target, distribute a wireless carrier, or provide a coherent reference, and each role exposes a different set of impairments. The integration architecture then reveals which component, signal, or resource is shared by communication and sensing. We therefore examine the evidence along the axes of physical platform and coupling mechanism.

Table IV reports the mutually exclusive family counts and corpus shares. Photonic terahertz and fiber together account for 125 studies, representing 60.7% of the corpus.

The family profiles below provide the corresponding physical paths, observables, coupling mechanisms, active constraints, transfer conditions, and evidence anchors. Each study

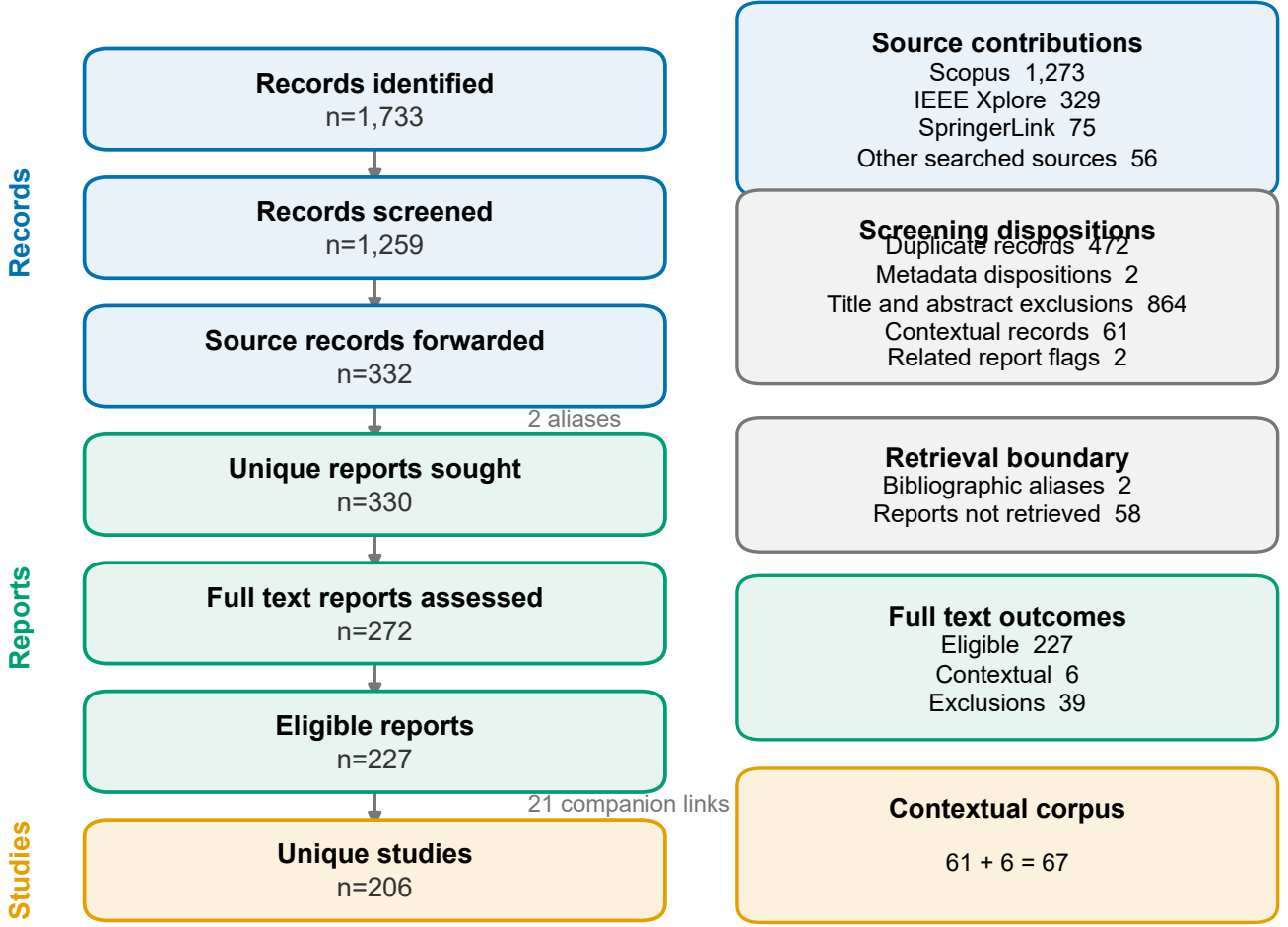


Fig. 3. Selection of the O-ISAC evidence base. Six sources yielded 1,733 records and 1,259 entered title and abstract screening. The retrieval route contained 330 unique reports after two bibliographic aliases were consolidated. Assessment of full texts covered 272 reports and produced 227 eligible reports. Companion mapping linked these reports to 206 unique studies. The contextual corpus contains 67 records that support field positioning.

TABLE IV
CANONICAL PHYSICAL ORGANIZATION OF THE CORPUS OF 206 STUDIES

Platform family	Studies, <i>n</i> (%)
Photonic terahertz	69 (33.5%)
Fiber	56 (27.2%)
Visible light communication and LiFi	38 (18.4%)
Free space optical	31 (15.0%)
Hybrid optical	9 (4.4%)
Other optical	3 (1.5%)
Total	206 (100.0%)

Note. Each study is counted once under its dominant platform family. Media and integration mechanisms can overlap within the represented systems. The counts summarize how the reviewed evidence is distributed across the six platform families.

belongs to one platform family, while its coupling mechanisms may overlap. Transmitters supplied through fiber enter the terahertz family when photonic generation and conversion create the wireless carrier directed at the target. Fiber remains the organizing platform when it transports signals to downstream radio frequency identification (RFID), radio, or radar sensing.

Hybrid membership begins when interaction between physical segments shapes the result. This organization locates the target path and guides the synthesis that follows.

A. Physical Platform Families

1) *Photonic Terahertz Systems:* Photonic terahertz is the largest family, comprising 69 of the 206 studies (33.5%). The optical chain generates, distributes, or processes a coherent carrier before conversion to a wireless terahertz or millimeter wave target path. It yields delay, range, Doppler, angle, imaging, and channel or target response observables. Communication and sensing are coupled through hardware, references, waveforms, beams, or resources [6], [7], [37]. This upstream sharing ties both functions to the same optical conditions. Laser coherence and linewidth govern reference stability, while component bandwidth, conversion fidelity, and efficiency shape the bandwidth, phase structure, and power delivered to the antenna [38], [39]. At the radio interface, antenna aperture, gain, and alignment govern launch and collection, molecular absorption shapes propagation, and target

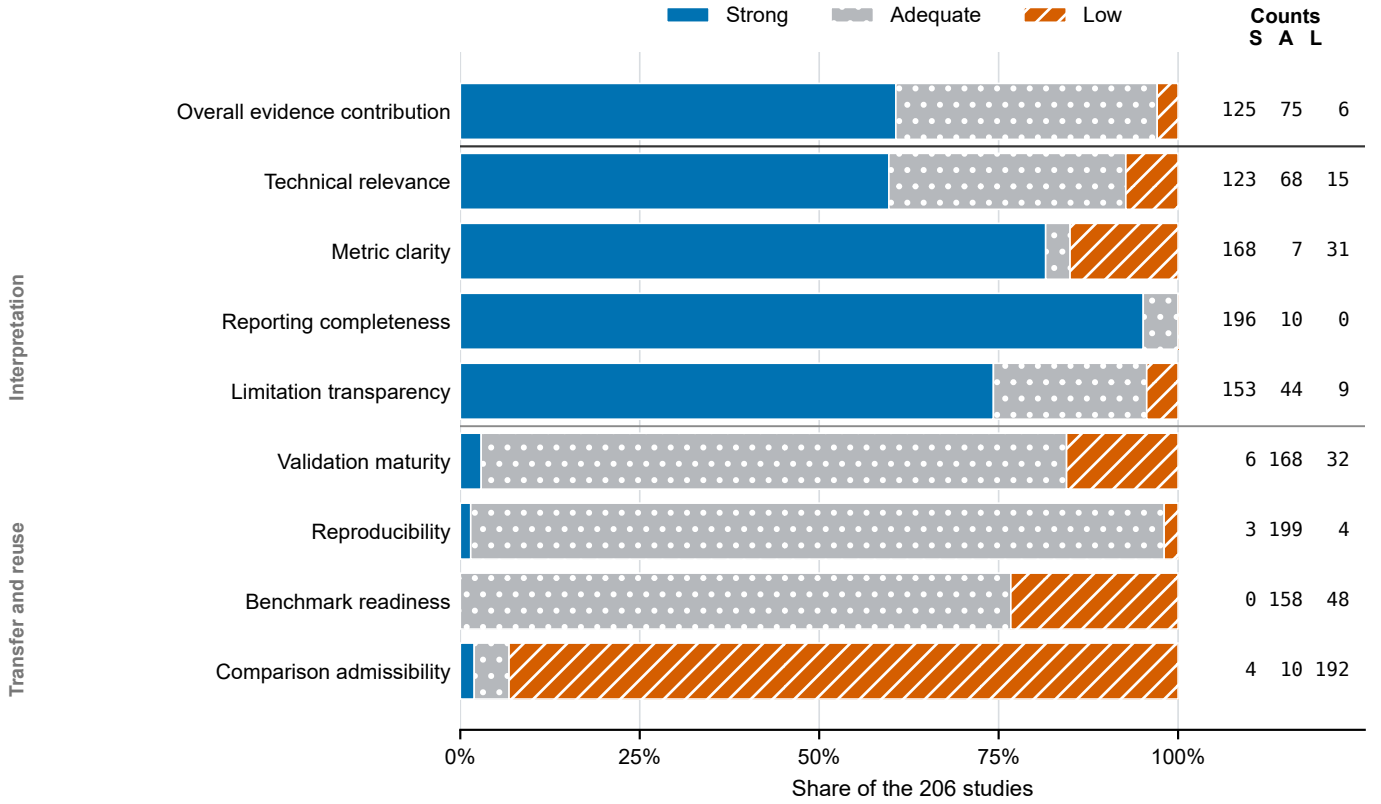


Fig. 4. Technical appraisal profile of the 206 included studies. Each row assigns all studies to strong, adequate, or low evidentiary support under the TQAF rules developed for this review. The interpretation group covers technical relevance, metric clarity, reporting completeness, and limitation transparency. The transfer and reuse group covers validation maturity, reproducibility, benchmark readiness, and comparison admissibility. Overall evidence contribution remains a separate summary category.

scattering forms the returning echo in radar configurations [40], [41].

Where the functions meet varies by architecture, with some systems dividing a multicarrier frame between sensing and communication and others placing the joint decision in a phased array controlled through photonics [6], [37]. Wavelength tuning and multibeam reception move coordination to the source or the receiving front end [42], [43]. At the waveform level, one multichannel design combines linear frequency modulation (LFM) with communication symbols across coherent carriers [38]. Other designs add source agility through frequency hopping or organize the signal over delay and Doppler through orthogonal multiplexing [41], [44]. Information can also close the loop, as burst SNR estimates guide both probabilistic shaping and ranging, while a simulated radar waveform carries communication data [40], [45]. Together, these designs show how the joint decision can move from hardware to waveform and to the information used by both tasks.

The location of that decision also determines how the measurement is formed. A returned echo may be remodulated onto light and carried back over fiber for centralized processing, where photonic dechirping expresses target delay at a lower frequency [41], [46]. Along this path, phase noise associated with laser linewidth degrades the recovered signal, while delay mismatch between optical and electrical paths

shifts the ranging observable [46]. Laboratory experiments reveal waveform, conversion, and beamforming behavior under controlled conditions [6], [38], whereas a 270 m outdoor link adds wireless propagation and operation of the complete chain [37]. Comparison across photonic terahertz systems depends on aligned optical and wireless stages, aperture and beam conditions, receiver processing, and validation setting.

2) *Fiber O-ISAC*: Fiber comprises 56 studies (27.2%) and follows two physical paths. The guided link may form the sensing medium, with disturbances observed through backscatter, carrier phase, modal coupling, or an inserted sensor [47], [48], [49], [50]. Fiber may instead transport signals to a downstream RFID, radio, or radar segment, so the target interaction occurs after optical distribution [51], [52], [53], [54], [55]. The corresponding observations include distributed events, vibration, strain, and environmental change in the guided path, together with downstream tag state, range, localization, and imaging [47], [51], [52]. The measurement boundary therefore lies either within the guided path or across the complete optical and wireless chain.

Coupling recurs through shared hardware, wavelengths, waveforms, links, or resource decisions. Integration can reuse the communication signal itself [48], [49], [56], [57], [58] or place distinct data and sensing signals on shared infrastructure [1], [47], [50], [59], [60]. The first form ties sensing to the symbols, carrier, phase, or supported modes, whereas the

second ties both functions to the same fiber and receiver conditions. Guided sensing is constrained by dynamic range, leakage, reflection, nonlinearity, reference stability, topology, and traffic, whereas transported sensing additionally depends on conversion, timing, remote calibration, and wireless geometry. Evidence comes from laboratory links, live access networks, and a deployed submarine route. For fiber evidence, comparison depends on the interrogation or transport role, topology, launch and receiver conditions, conversion and clocking, target path, traffic, and validation setting [47], [51], [52], [55], [56], [57], [60].

3) *Visible Light Communication and LiFi Systems*: Visible light communication and LiFi comprise 38 studies (18.4%) and place communication and sensing within the same illuminated scene. The grouping preserves their distinct meanings at the link and networking levels. Direct or reflected room and device paths through emitters, cameras, and photodiodes support position, pose, distance, imaging, gesture, recognition, and physiological sensing [2], [61], [62]. Cameras and photodiodes expose different balances between spatial detail and reception speed, while nonnegativity, dimming, flicker, source linearity, ambient light, field of view, and exposure constrain what can be communicated and observed [61], [63], [64], [65]. Coupling recurs through hardware, waveform, geometry, resource decisions, and the application loop. It may arise through a shared observation [62], [66], through geometry and resource decisions [67], [68], [69], [70], [71], or through integrated emitters and detectors [72], [73], [74].

Illumination is the common comparison boundary across these designs. Geometry and multipath shape localization, reflected observations support gesture and recognition, and learned models extend the same optical field to health sensing tasks [61], [65], [69], [70], [75], [76]. Analytical and simulated studies isolate geometry and resource choices, whereas prototypes and field experiments add lighting, receiver, and control conditions [61], [66], [67], [69], [71], [73]. Visible light comparisons depend on matched task definitions, layout, orientation, illumination, detector, processing, and validation conditions.

4) *Free Space Optical Systems*: Free space optical systems comprise 31 studies (15.0%) and use a directional link, optionally combined with target illumination and return, over direct or reflected aperture and beam paths. These paths support ranging, localization, imaging, tracking, and atmospheric or channel observation [77], [78], [79]. Geometry is therefore part of both the link and the measurement. Coupling recurs through the link, waveform, aperture or beam, and spatial or temporal resources, while pointing, divergence, aperture, reflectivity, loss, turbulence, and motion constrain the resulting evidence. Direct, reflected, scanned, and steered paths illustrate these relations [77], [80], [81], [82]. Retroreflection and multicarrier reception connect returned power to range and position [78], [79]. Structured waveforms and steering apertures place coupling in the signal or beam [83], [84], [85], [86], while atmospheric estimates can guide beam shaping or link selection [3], [87].

Across these forms, beam geometry determines both spatial selectivity and the power collected at the receiver. Range,

angle, and image features therefore remain tied to calibration, aperture, pointing, and motion [77], [78], [79], [80], [81]. Controlled links reveal waveform and estimator behavior, whereas operational ranging adds atmospheric variation and continuous field observations [3], [79]. Comparison across free space optical systems depends on matched geometry, calibration, target properties, weather, motion, and validation conditions. Hybrid systems extend that comparison across interfaces between physical segments.

5) *Hybrid Optical Systems*: Hybrid optical systems comprise nine studies (4.4%) in which consequential interaction occurs among optical, fiber, free space, or radio segments and the perturbation and observation may straddle a conversion boundary. Their observables include range, vibration, and target or network state within individual segments, while coupling arises through a central reference and through links, resources, control, or applications shared across segments. Conversion loss, segment power, synchronization, calibration, and timing are therefore active constraints [34], [88], [89]. Hybrid comparisons trace every segment, conversion, clock, and measurement plane together with the validation of both functions.

Unlike fiber used only for downstream transport, hybrid membership requires conversion or feedback across segments to shape the result. This occurs when interaction between fiber and wireless segments is consequential [34], [88], when a shared photonic source spans fiber, free space, and millimeter wave stages [90], or when sensing in one segment guides service in another [89], [91]. Other designs establish a common signal origin through shared time modes or a photonic waveform with frequency agility [92], [93].

6) *Other Optical Systems*: The three other optical studies (1.5%) use distinct mechanisms based on optical control or coherent carriers. These comprise a microwave field generated by a microcomb, a solar blind ultraviolet path, and a quantum optical path. Their observables are, respectively, a field pattern, particulate or smoke response, and phase [94], [95], [96]. Coupling is specific to specialized hardware, reference, and application choices, and each case is bounded by its own propagation, target, receiver, and measurement models. Each numerical result is therefore interpreted within the native mechanism and measurement model of its system. Together, the hybrid and other optical cases complete the physical map before the analysis turns to coupling mechanisms shared across families.

B. Integration Architectures and Resource Coupling

Across these physical families, integration architecture locates where communication and sensing meet along the target path. The coupling may enter through shared hardware, a common carrier or waveform, coordinated resources, a propagation link, or a service decision. Several mechanisms can coexist in one system. Each label identifies one coupling location while preserving the rest of the chain for separate interpretation.

Figure 5 locates seven recurring coupling mechanisms along a generic signal and service path. A study may occupy several locations, so the badges report overlapping coverage across the corpus.

Coupling locations may overlap within one study

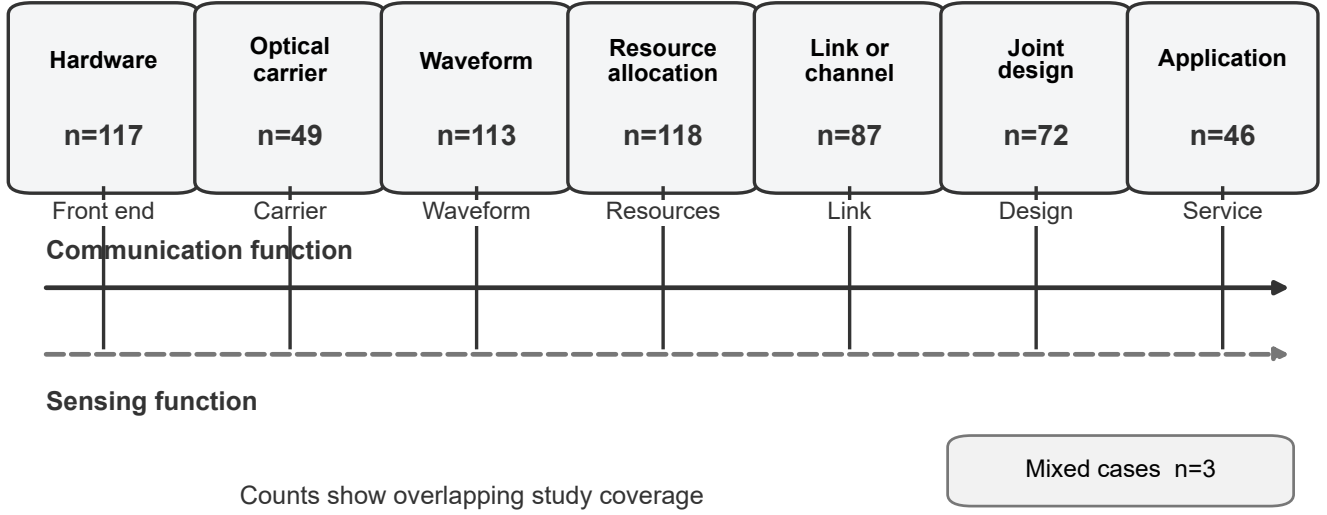


Fig. 5. Locations of communication and sensing coupling in the corpus of 206 studies. Equal cards identify hardware, optical carrier, waveform, resource, link, joint design, and application coupling along the system path. The badges report overlapping study coverage, with three mixed boundary cases shown separately. Card position identifies the typical coupling location, while each card retains equal visual weight.

Resource, hardware, and waveform sharing can form one connected design chain. Application coupling acts at the service level and carries a different engineering meaning. The following subsections trace these locations from physical reuse to coordinated resources and design.

1) *Shared Hardware, Carrier, and Link:* Hardware, carrier, and link sharing describe different physical relations. Reusing a component makes its bandwidth, coherence, dynamic range, or alignment conditions relevant to both functions. A common optical carrier adds a phase or wavelength relation, while a common link adds shared propagation and calibration conditions. Together, these relations show which design choice is common and which constraint follows it.

The physical path gives reuse its engineering meaning. In fiber, traffic and sensing share the guided structure, with a resonance shift, phase, or backscatter conveying the sensed change [1], [97]. A retroreflective free space path supports information delivery together with ranging or localization [77]. Arrays controlled through photonics instead share optical generation and beam control while the target path remains wireless [6]. Comparison therefore follows the component role, carrier relation, propagation path, and calibration.

2) *Shared Waveforms:* Waveform sharing begins when one signal structure contributes to both decoding and sensing. The shared element may be the complete waveform or a selected pilot, chirp, code, polarization state, or spatial pattern. This common design links payload efficiency, sensing observability, and receiver processing. Structural sharing records that relationship. Concurrent performance adds the requirement that both outcomes be observed at a matched operating point, a distinction carried into Section V.

The evidence illustrates several forms. Pilots inserted for channel estimation can also provide a sensing reference [98]. One photonic terahertz prototype generates a waveform whose frequency steps provide the radar dimension and whose amplitude carries data. Its experimental evidence is strongest for waveform generation and communication response [99]. A visible light design combines temporal data with spatially structured illumination and uses localization to adapt beam-forming [2]. The shared structure is similar, while its physical and receiver implementation remains specific to the platform.

3) *Resource Allocation and Joint Design:* Resource coupling coordinates temporal, spectral, power, spatial, or processing degrees of freedom. An allocation can change payload opportunity, link margin, sensing observability, geometry, or latency. Joint design connects the two functions more directly by choosing a variable, objective, or constraint in view of both outcomes. The resource, allocation rule, and operating conditions therefore define the resulting evidence.

A visible light model for vehicle to everything communication makes exposure time a shared decision variable. Its coexistence result is analytical and numerical, while laboratory communication and sensing gains were obtained in separate configurations [63]. In one reported terahertz configuration, a rapidly swept photonic source and common waveform plan support simultaneous data transmission and ranging [7]. These cases distinguish a modeled optimum from a measured concurrent configuration. Section V therefore reads each relationship within its operating region and evidence form.

4) *Coupling at the Application Level and Boundary Cases:* Application coupling arises when sensing informs beam or link selection, trajectory planning, blockage response, or network

control. The shared element is then the system action, even when the signal paths remain separate. A near infrared wearable combines communication and physiological sensing on a common material platform while retaining separate detector roles [74]. A hybrid last mile design places free space optical delivery and fiber Bragg grating monitoring within one system model [89]. Their coordination lies in hardware and service context.

Three studies cross temporal or physical segment boundaries. One performs localization and imaging before data delivery on a shared optical source and path with separate receivers [80]. Two use sensed free space optical channel state to adapt trajectory and resource allocation or cell placement and user association across optical backhaul and radio access [100], [101]. The mixed label preserves this path and keeps each result tied to the segment, function, and operating stage that produces it.

C. Interpretation Across Platforms

Read together, the platform and architecture views form one mechanism map. The platform establishes the physical transfer conditions, while the architecture identifies the shared choice. In fiber, waveform sharing inherits guided propagation and reflections, together with launch and receiver conditions [56], [57]. In free space optical systems, the same mechanism is shaped by aperture, geometry, and alignment, whereas terahertz systems assisted by photonics also introduce conversion from optics to radio [6], [77]. Application coupling can instead coordinate separate optical and radio paths through a common service decision [100].

Across these mechanisms, sharing concentrates design around a common component, reference, or resource. That choice can couple the functions through power balance and interference, while synchronization and receiver dynamic range constrain their joint operation [1], [34]. Section V therefore examines which communication and sensing outcomes change together at a matched operating point and which conditions explain that relationship.

V. PERFORMANCE METRICS AND JOINT DESIGN TRADEOFFS

After locating a shared choice, the analysis turns to the reported communication and sensing outcomes. A reported value gains engineering meaning from its task, its observation point in the signal chain, and its operating and validation conditions. The comparison profile connects these elements to the performance evidence.

The primary synthesis contains 4,779 metric records. Each record is an extraction unit that retains the reported quantity together with the definition and source conditions available for it. The denominator therefore represents reported quantities in their original contexts. Data rate, bit error rate (BER), and range resolution have the widest study coverage, appearing in 85, 60, and 50 studies, respectively. Since one study may report several families, these totals provide overlapping views of the evidence.

Beyond these familiar measures, the evidence extends from link and waveform behavior to outcomes observed at the device, estimator, application, and implementation levels. This breadth makes semantic alignment the organizing principle for the analysis that follows.

A. Applying the Comparison Profile to Individual Metrics

Applying the comparison profile defined in Section II, each metric row was coded by source definition, task, measurement plane, operating point, validation method, and value provenance. Communication and sensing results form a joint operating point when the source links them to the same condition set. Table V summarizes domain coverage and the analytical role assigned to each record.

TABLE V
METRIC EVIDENCE PROFILE FOR 4,779 PRIMARY RECORDS

<i>Metric domains</i>		
Domain	Records, <i>n</i> (%)	Studies, <i>n</i> (%)
Sensing	1,816 (38.0%)	203 (98.5%)
Communication	1,328 (27.8%)	194 (94.2%)
Joint	870 (18.2%)	158 (76.7%)
Implementation	476 (10.0%)	64 (31.1%)
Smaller normalized domains	289 (6.0%)	33 (16.0%)
<i>Analytical roles</i>		
Role	Records, <i>n</i> (%)	Use in the survey
Bounded relation	118 (2.5%)	Conditional comparison across studies
Native context	4,661 (97.5%)	Mechanism, feasibility, and qualitative synthesis

Note. Domain rows and analytical roles each partition the 4,779 primary records, while study counts overlap. The complete metric register contains 4,861 rows. The additional rows comprise 31 contextual records and 51 records documenting source conflicts. Comparison follows the reported definition, task, measurement plane, conditions, and validation. Electronic Supplement S-Evidence provides the complete records and provenance, including quantities grouped under the broader other category.

The domain distribution shows where metric evidence is concentrated, while the analytical role determines how each record enters the survey. The following subsections explain how task, measurement plane, and operating conditions shape communication, sensing, joint, and implementation metrics.

B. Metrics Across Functions

1) *Communication Metrics:* A communication rate gains meaning from what the source counts. Symbol rate tracks signaling pace, while data rate reflects the payload and overhead included in that count. Throughput describes delivered service, capacity provides a model bound, and spectral efficiency relates useful data to its stated denominator. In a terahertz prototype enabled by photonics, net rates follow communication subcarrier allocation, payload occupancy, and coding overhead [37]. An embedded pilot orthogonal frequency division multiplexing (OFDM) simulation defines spectral efficiency as the data subcarrier fraction and reports cyclic prefix overhead separately [98].

Reliability shifts attention to the event and its location in the signal chain. BER measured before and after forward error correction describes two stages of the link, while packet loss and outage describe broader delivery boundaries. A fiber experiment over a long span reports receiver OSNR and BER before correction during concurrent distributed sensing [102]. A robotics study reports packet loss over distance and view angle, display delay over queue size, and formation control under dynamic motion as separate results [61]. The reported reliability value is therefore tied to its event, observation point, time scale, and system boundary.

The observation point also determines how signal quality and power are interpreted. OSNR relates signal power to noise in a stated optical reference bandwidth, while electrical SNR follows photodetection and digital SNR follows equalization. Launch and received power likewise refer to the transmitter and receiver ends of the link. A fiber polarization sensing experiment traces recovered vibration SNR across an OSNR sweep, connecting the sensing response to a stated optical condition [32].

Distance and latency complete the profile by locating the result in space and time. Fiber span and wireless separation describe link geometry, while target range and coverage radius describe the observed or served region. Latency may include propagation and buffering, signal acquisition and processing, or the full application loop. Stated endpoints, included stages, and operating conditions define the scope of each value. The complete profile connects each communication metric to the system behavior it represents.

2) *Sensing Metrics*: Sensing metrics first take meaning from the task and quantity they represent. Range and position estimates, motion measurements, images, and class decisions refer to distinct observations and references. Resolution describes separability along a stated dimension, while estimation error relates an estimate to a reference. Repeatability captures variation across trials, and a model bound supplies a conditional benchmark. These distinctions establish the comparison basis.

Representative studies show how this contract works. A fiber reflectometry system reports breakpoint localization error and range resolution separately for laboratory and simulated conditions [103]. A visible light localization experiment distinguishes lateral estimation error from the smallest demonstrated displacement and summarizes signed displacement with means and standard deviations across repeated trials [2]. An optical phased array simulation relates imaging root mean square error to electrical SNR under a defined scene, waveform, and beamforming constraint [33]. Each result retains its summary, geometry, and validation setting.

Detection and classification add dataset and decision context. Their metrics take meaning from class balance, decision thresholds, background conditions, and the relation between training and test data. CORE-Lens reports recognition accuracy and optical camera communication BER across stated class, data rate, and background settings, with runtime evaluated separately [104]. VehicleTalk evaluates link performance, distance estimation error, and vehicle status recognition under separate prototype conditions [62]. Their application meaning

follows from the attached classes, geometries, and test protocols.

Physical quantity sensing follows the same logic by linking each measurand to a reference and calibration. Spatial support and acquisition time locate the observation, while the aggregation rule shapes its summary. The resulting profile connects the task, reference, geometry, processing, and validation setting to the reported sensing outcome. Together, these elements define the conditions for comparison.

3) *Joint and Implementation Metrics*: Joint metrics connect communication and sensing through a shared operating point or resource rule. A matched tuple uses the same conditions for both functions. A shared resource can also link values that retain their own reported conditions. In an embedded pilot OFDM simulation, BER and mean absolute pilot error are reported at the same SNR with eight pilots, while spectral efficiency follows the corresponding data subcarrier fraction [98]. The pilot error measures deviation of known pilot symbols in the simulated channel. Together, the three values show how pilot allocation connects communication reliability, resource efficiency, and the reported sensing measure.

Optimization creates another form of joint evidence by selecting an operating point through shared design variables. In a simulated optical phased array system, precoding matrices and photodiode orientations are selected together. The design maximizes a sensing contrast metric under a communication quality threshold and emitter power constraints [33]. Geometry, constraint thresholds, and metric definitions define the scope of the solution.

Implementation metrics place these operating points within the platform and application chain. Receiver bandwidth, converter rate, memory, acquisition time, and latency show where processing burdens arise. An outdoor photonics assisted terahertz prototype evaluates receiver pipeline latency in hardware analysis and communication and ranging performance in a field experiment [37]. A smartphone camera prototype evaluates recognition and communication in the laboratory and benchmarks module latency, memory, and complete cycle timing on the device [104]. Together, these measures connect joint performance to the processing chain and locate the effects of shared design choices in system behavior.

C. How Communication and Sensing Performance Interact

Communication and sensing often share bandwidth, power, waveform components, hardware, and processing. A design change in any of these elements can move both functions to a new operating point. More pilot symbols may strengthen a sensing reference while leaving fewer resources for data. Wider bandwidth may support both a higher data rate and finer range resolution until a receiver or platform limit becomes active. The evidence is organized by the shared choice that produces each pattern.

The synthesis covers 402 relationships from 168 studies. Figure 6 separates record type, overlapping study coverage, and condition dependence across 11 recurring mechanism families. Each relationship remains tied to its waveform, geometry, target, observation point, and validation setting.

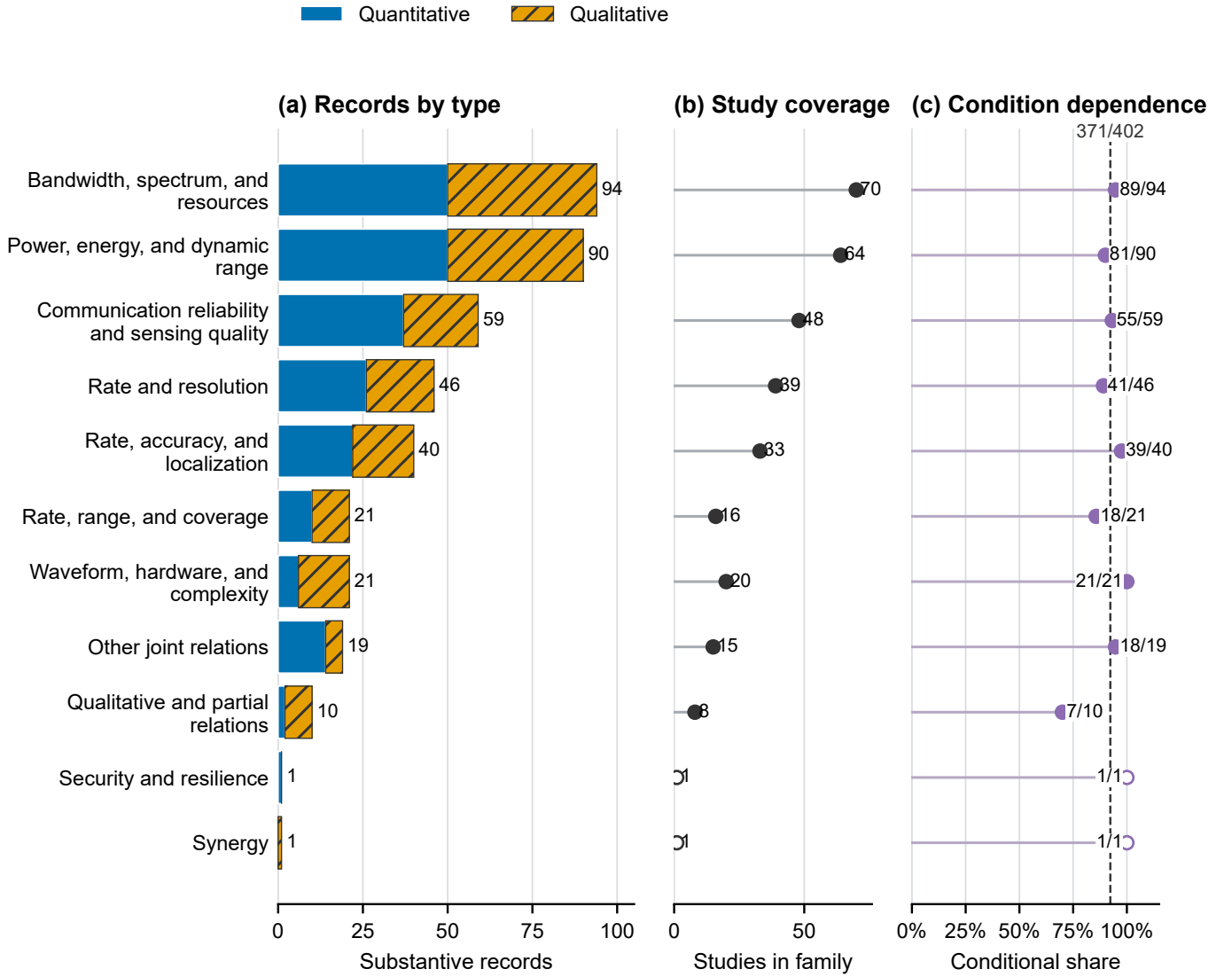


Fig. 6. Evidence profile for 402 substantive joint performance relationships from 168 studies. Panel (a) separates 218 quantitative and 184 qualitative records. Panel (b) reports overlapping study coverage across the mechanism families. Panel (c) reports the share of records whose interpretation depends on stated conditions, with 371 such relationships across the profile. Family study counts may overlap, and frequency describes corpus coverage.

Resource and power mechanisms carry much of the coded evidence, while condition dependence remains common across the families. This pattern directs the discussion toward the shared mechanism and the operating condition that shapes each reported relation.

1) *Bandwidth, Spectrum, and Resource Allocation:* Bandwidth links the two functions when both draw on the same signal space. That space may be divided across subcarriers, wavelengths, guard intervals, observation time, or the usable response of a device. Embedded pilots show the basic pattern. Increasing pilot support improves the reported pilot deviation proxy while changing BER and the data carrying fraction [98]. Protection intervals and chirp sweeps create related exchanges by improving sensing isolation or pulse compression within device bandwidth and subcarrier limits [34], [59]. In each case, the effect follows the resource being shared and the limit that becomes active.

Waveform structure determines how the shared signal space

is used, with format and operating mode setting time and spectral sharing [7], [77], [80]. Within that structure, digital designs assign subbands, chirp parameters, power, and weighted resources [105], [106], [107], [108], [109]. Pilot strength, placement, and synchronization fields define reference support and communication margin, while receiver partitioning and ranging sequence redundancy set the usable signal region [37], [110], [111], [112], [113], [114]. A tunable laser study supplies qualitative generation context for the same design space [115]. Together, the allocation rule and receiver condition define the operating point reported in each design.

Resource sharing also appears in time. Observation duration, camera exposure, and task scheduling connect sensing quality and cadence with communication goodput or execution time [63], [104], [116]. Time division, chirp repetition, coding traffic, security control, and embedded pulse count place both functions within a shared frame [62], [66], [117], [118], [119].

Other designs reserve spectral, temporal, spatial, polarization, or phase structure to manage interference, coverage, and receiver complexity [5], [43], [54], [120], [121], [122]. The observed outcome therefore depends on the frame period, exposure response, isolation rule, and remaining occupancy.

When both functions occupy the same waveform, bandwidth affects pulse compression, estimation precision, sidelobes, and radar image quality together with communication support [123], [124], [125], [126], [127]. Stimulated Brillouin scattering, frequency partitioning, and constellation choice reveal device and spectral limits [38], [128], [129]. Doppler refinement depends on phase noise and computation [130]. A communication and sensing bandwidth ratio gains meaning from total bandwidth, target range, and receiver architecture [131]. These results show that signal bandwidth and receiver processing jointly define the operating point.

Reuse extends this mechanism by carrying sensing content in a shared waveform or an existing transport path [39], [132], [133]. Phase coding, carrier grouping, radio frequency bandwidth, and space and wavelength allocation then set the occupancy and isolation available to each function [65], [134], [135], [136]. The result depends on how the receiver separates the shared content and recovers the underlying sequence.

Hardware and processing resources complete the picture. Reference recovery and coding consume timing or redundancy [79], [137], [138], while emitter bandwidth, source coherence, optical budget, and distributed learning traffic set hardware and network capacity [139], [140], [141], [142]. Receiver count introduces alignment and crosstalk constraints. In the reported prototype, four surrounding receivers were designed and one receiver was operated [143]. Phase control and observation settings affect the recoverable path [3], [144], while modulation, sideband, and time bandwidth controls affect signal recovery [145], [146], [147]. The same nominal allocation can therefore produce a different result when the platform or receiver changes.

2) *Power, Energy, and Dynamic Range*: Power affects both functions through the finite operating range of the transmitter, channel, and receiver. Launch and received power, carrier and probe ratios, device bias, modulation swing, and converter headroom refer to different points in that chain. As power changes, performance follows the component that first approaches a noise, nonlinear, or saturation limit. The reference plane and limiting component are therefore part of the result.

A fiber probe sweep shows this response directly. Raising the sensing to telecom power ratio improves sensing SNR and raises BER, with the strongest link penalty near the probe [148]. In a visible light prototype, one optical branch supports communication and ranging while another supports energy harvesting. Their allocation governs rate, ranging quality, and harvested power under separate receiver and endpoint bias conditions [149]. Similar ratios connect BER or OSNR margin with phase, polarization, and vibration observability in fiber [1], [32], [150], [151], [152]. In photonic wireless links, carrier and waveform ratios connect phase recovery, BER or error vector magnitude (EVM), and radar or ranging quality within the available receiver headroom [46], [153], [154], [155]. Comparisons are most informative when they use the same

power reference and receiver margin.

Absolute power sweeps reveal where the operating range ends. In fiber, launch and received power expose nonlinear interaction and sensing reach [102], [156], [157]. Photonic front ends examine operating limits associated with input power, modulator drive, converter linearity, and amplifier gain [52], [158], [159], [160]. Architecture then decides where the available power is used. Polarization paths, receiver taps, and branches assigned to individual functions distribute power before estimation [161], [162], [163]. Propagation, insertion, and alignment losses determine how much reaches those branches [60], [81], [97]. The remaining margin sets the feasible operation of optical phased arrays, communication users, sensing estimators, and illumination services [33], [82], [164], [165].

Power allocation can also be written into the waveform. Service branches, reference copies, and relative waveform amplitudes connect radar masking, ranging, data delivery, and energy harvesting through a shared allocation [166], [167], [168], [169]. Sequence transitions, coding choices, and chaotic source parameters change how communication and sensing information is recovered [170], [171], [172]. Frequency separation, probabilistic shaping, sensing slots, and carrier reuse move energy and occupancy across signal dimensions [173], [174], [175], [176]. The waveform definition, hardware margin, and performance threshold determine the resulting operating point. These same factors also govern reliability and sensing quality.

3) *Communication Reliability and Sensing Quality*: Communication reliability and sensing quality become linked after the waveform and receiver are selected. Communication evidence uses BER, EVM, quality factor, and required OSNR, while sensing evidence uses SNR, estimation error, and detection. These measures can move together as leakage, nonlinear mixing, phase noise, cancellation quality, or receiver competition changes. Some controls reduce an interference path, while others redistribute energy within the shared signal. This distinction explains why similar metric pairs can follow different trends.

Two studies show why interference control and power redistribution require separate interpretation. In a fiber link combining communication with coherent reflectometry, sequence shaping reduces nonlinear interference induced by signal transitions. This lowers the communication penalty while retaining phase sensing at the reported condition [170]. In a millimeter wave waveform, a direct current offset strengthens the sensing component and raises the radar signal to interference ratio together with communication EVM [177]. The first mechanism changes how interference is produced, while the second redistributes energy within the signal. Their trends therefore arise from different controls.

Architecture then determines where these effects enter the chain. Shared path tests evaluate communication reliability beside sensing outcomes at stated operating points and validation settings [103], [178], [179], [180]. Dedicated monitoring wavelengths and reused channel estimates locate sensing support in optical filtering or receiver processing, with communication performance described qualitatively [47], [181]. The

path and receiver therefore define the scope of each reliability result.

Once the architecture is fixed, waveform and inference choices determine the reported communication and sensing metrics. Carrier choice, direct current offset, and phase modulation affect recovery at stated settings [99], [182], [183]. Coding, guard structure, and waveform composition affect BER, correlation quality, and ranging error [184], [185], [186]. At the inference stage, loss weighting connects communication reliability with distance and velocity errors [187], while Kalman tuning acts within the sensing branch [188]. Shared sequence structure and finite block phase processing place radar detection and phase estimation under their sample and noise models [53], [96]. Phase index and pulse position timing likewise connect communication error with radar SNR or time of arrival precision [189], [190]. The sample, noise, timing, and calibration definitions give each result its scope.

At the system level, access placement guided by channel sensing connects link capacity with weather and demand [101]. The aggregation of sensing traffic connects capacity with target count and resolution requirements [55]. Simultaneous use of time and frequency creates direct waveform coupling [191], while shared hardware with waveform switching creates a different path [192]. Waveform robustness under fading and learned phase sensing place the limiting response in the channel or processing stage [193], [194]. These studies connect reliability and sensing quality to the complete waveform, receiver, channel, and processing chain. The same chain also governs rate, resolution, accuracy, and range.

4) *Rate, Resolution, Accuracy, and Range:* The link between communication rate and sensing begins with the design variable that changes the rate. Bandwidth and symbol timing can alter resolution directly, while frame structure, geometry, and processing change the resources available to the estimator. The controlling variable and its operating conditions explain why a similar rate change can produce different sensing outcomes.

Waveform parameters provide the clearest starting point. In optoelectronic oscillator and remote ranging designs, code length and period connect bit rate or capacity with distance resolution and unambiguous range [195], [196]. Power division changes OFDM reliability, reach, and ranging error in a retroreflective prototype, while image decomposition links stripe recovery with scene reconstruction and recognition [197], [198]. LFM slope indexing and time division designs in the W band place their controlling choices in slope selection and frame duration [199], [200]. The shared parameter sets the direction of the relationship in each case.

Frame structure creates a second path by deciding when and where each function receives signal support. Dynamic allocation and intensity scaling distribute support across payload, sensing patterns, and coverage [2], [201]. Orthogonal time frequency space parameterization and complete grid OFDM reuse place both functions in a common time frequency structure [88], [202]. Separate measurement paths and mode switching show how a similar waveform can support different forms of integration [203], [204]. The frame and measurement schedule are therefore part of the result.

Geometry sets the region in which those resources remain useful. Steering and access point placement define communication and localization coverage, while surface allocation and blockage aware access select the paths available to each function [6], [91], [205], [206]. Source directivity and tag delay spacing further affect coverage, range separation, and concurrent return recovery [207], [208]. The reported layout is therefore part of the observed trend.

Signal construction can tie payload more closely to sensing observability. Chip length changes occupied symbol rate together with ranging precision and security tolerance, while embedded communication data affects distributed sensing fidelity [56], [209]. Amplitude shift keying rate and guard duration connect communication timing with range resolution and ambiguity [210], [211]. Waveform switching provides a related parameter coupling, with rate and resolution evaluated through separate parameter changes [212]. In the terahertz system, the available device band supports quadrature phase shift keying (QPSK) communication and the LFM bandwidth used to achieve range resolution [213]. The waveform and hardware band together define the feasible region.

Optimization studies express these mechanisms as operating regions. Energy and asymptotic block length define communication and detection rate pairs, while nonorthogonal multiple access power allocation connects aggregate rate with ranging error [64], [214]. Other studies combine capacity with ranging information or compare separate and joint operating configurations [90], [215]. Probabilistic shaping and embedded data create further regions through information rate, sidelobe level, and estimation error [216], [217]. The selected operating point follows the stated objective, constraints, and normalization.

Processing adds another layer because the estimator decides how much of the available signal becomes useful sensing information. Sampling period, spectral weighting, zero padding, and transform length exchange response time and computation for resolution under stated acquisition settings [218], [219], [220], [221]. Rate selection can tune the sensitivity of a density estimate derived from BER [222]. In a separate application, classifier size changes recognition accuracy and edge processing burden [76]. These studies distinguish waveform rate from the update and computation needed to interpret it.

Across the network, the relationship moves from a single waveform to the traffic and spectrum available across the system. Transport capacity controls how many sensing streams meet an echo quality threshold [223]. Separated bands, idle time frequency cells, cyclic prefix correlation, and shared optical waveforms connect occupied spectrum with resolution or estimation support [224], [225], [226], [227]. A visible light study carries this logic into a proposed capacity and localization problem [228]. The evidence thus distinguishes observed operation from a proposed design space.

5) *Waveform, Hardware, and Complexity:* Implementation complexity depends on where a design places its burden. Source coherence, receiver structure, calibration, synchronization, computation, memory, and latency represent different costs. Reusing a component may move work into processing or control, so implementation complexity must be assessed across the complete chain.

Two photonic systems show how this transfer occurs. A terahertz design uses coherent comb generation to stabilize the QPSK and LFM paths and concentrates complexity in the optical source chain [213]. A fiber experiment over a long span uses lower cost lasers with broad linewidths and recovers coherent data and vibration through residual carrier processing [229]. Each architecture preserves performance through a different balance of source hardware and digital processing.

6) *Other Implementation Relationships*: Other implementation relationships show how a shared choice appears at different layers of the system. Its effect may emerge as protection, joint gain, device sensitivity, estimator burden, control freshness, or allocation latency. Each case carries its own conditions and prepares the comparison limits that follow.

Security and synergy illustrate two possible directions. Phase keyed recovery uses ranging tolerance to create a physically bounded decryption region [209]. With spatial alignment, communication gain and target profile estimation can benefit from the same optical design [82]. Their geometry, objective, and validation setting explain why one relationship expresses protection and the other expresses joint gain.

Other cases locate the relationship within the implementation chain. Electrode transparency connects device architecture with photodetector sensitivity, while passive return loss and repeater structure affect access for fault localization [74], [230]. Capacity and distance estimation define a model operating region [231]. Queue freshness enters the robotic control loop, and local spectral scanning connects computation with range resolution [61], [232]. Weighted priorities guide resource allocation, while orthogonal task channels shape application latency [233], [234]. Read together, these cases locate the relationship at the device, estimator, control, or allocation layer and provide the conditions needed for comparison across studies.

D. Conditions for Comparison Across Studies

Comparison across studies begins with alignment. The bounded relations summarized in Table V connect studies when their task, metric definition, measurement plane, operating point, and validation setting can be matched through the framework in Section II. Other metric evidence explains mechanisms, feasibility, and behavior within individual systems. This separation keeps each reported value attached to its original engineering meaning.

Tradeoff evidence follows the same principle. Because objectives and constraints vary across systems, the synthesis compares the shared mechanism and the conditions that produce the observed response. This approach shows why a relationship recurs and when it can guide another design. Its engineering value also depends on the validation setting and on how completely the configuration can be reconstructed. Section VI examines these requirements through validation settings, field evidence across both functional domains, and reconstructability.

VI. VALIDATION EVIDENCE, RECONSTRUCTABILITY, AND BENCHMARK READINESS

Section V showed that comparison depends on alignment across the task, measurement plane, operating point, and validation setting. This section examines how far those operating points were tested and how completely their configurations can be reconstructed. It then relates validation settings, evidence from both functions, and artifact access to benchmark readiness.

A. Validation Settings and Methods

Figure 7 places each study's highest reported setting beside the validation methods used across the corpus. The first panel assigns one setting to every study, while the second allows several methods to contribute to one evidence chain.

The profile shows a broad laboratory and prototype base together with a smaller field subset. Analytical, simulation, laboratory, and prototype methods can contribute to one study, so the configuration trail across methods shapes interpretation.

These settings expose different parts of an O-ISAC system. Simulation studies isolate waveform and device sensitivities under controlled conditions [98], [202]. Laboratory and prototype systems reveal conversion, synchronization, and receiver effects in physical implementations [74], [88], [123]. Field studies extend the evidence to access networks, deployed fiber, submarine monitoring, and outdoor wireless links [37], [47], [51], [52], [136].

The highest setting describes the reach of the evidence. Its scientific meaning still depends on the claim and the test design. A controlled prototype may explore a wider parameter range than a field trial, while a laboratory experiment may reveal communication and sensing interactions more directly than a field dataset collected for one outcome. The setting therefore gains meaning when it is read together with the tested conditions, available controls, and functional coverage.

Each method serves a different analytical purpose. Analysis clarifies assumptions, simulation supports controlled parameter studies, and laboratory work reveals the behavior of physical components. Prototypes bring subsystems together, while field studies introduce at least one operational boundary. Their combined value grows when the configuration and conditions remain traceable as the evidence moves from one method to another.

Section IV shows that realistic validation conditions change with the platform. Fiber studies depend on topology, traffic, launch and receiver conditions, and the disturbance context. Visible light communication and LiFi studies add layout, orientation, illumination, detector, and processing conditions. Free space optical studies require geometry, pointing, target properties, weather, and motion. Terahertz systems enabled by photonics must also trace optical conversion, antenna alignment, and wireless propagation. A useful validation record therefore identifies the environment, measurement plane, target or disturbance, reference method, and exercised function.

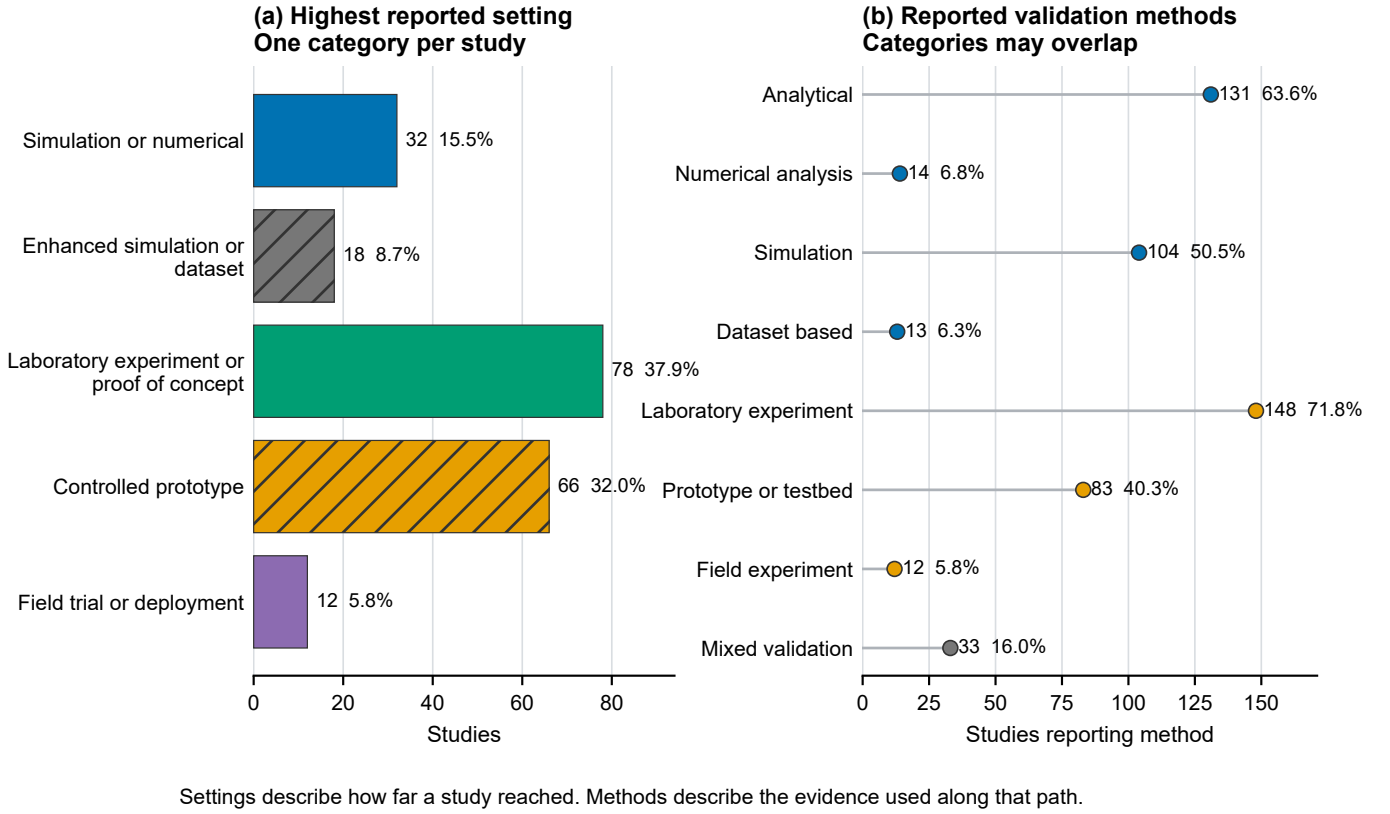


Fig. 7. Validation settings and methods in the 206 studies. Panel (a) assigns each study to one highest reported setting. Panel (b) reports analytical and numerical methods, simulations, methods based on datasets, laboratory experiments, prototypes, field studies, and mixed methods. Method categories may overlap because several forms of evidence can contribute to one study. The panels describe the reach and composition of validation as distinct properties.

B. Field Evidence Across Communication and Sensing

The highest validation setting and the coverage of both functions answer related questions. For 12 studies, a field trial or deployment was the strongest reported setting. Six of them also received TQAF validation score 3 because their records report field or deployment outcomes in each functional domain. The assessment is applied to each study. Electronic Supplement S-Appraisal reports the corresponding scores. Electronic Supplement S7, the paired function validation register, identifies the 12 studies and the subset of six.

The six studies cover several settings. They include terahertz and cellular links enabled by photonics, fiber access integrated with radio frequency identification, and space laser ranging and communication. They also include multicore fiber deployed in the field and submarine cable monitoring [37], [47], [51], [52], [79], [136]. Their diversity shows that field evidence for both functional domains appears across several platforms. Platform comparisons still follow the native validation conditions described in Section IV.

Electronic Supplement S7 records the available relationship timing and separate source locators for the two functions. The six cases establish field or deployment outcomes in both functional domains at the level of each study. Concurrent joint operation is established when both outcomes are traceable to the same configuration, time interval, and condition set.

The distinction becomes important when evidence is produced at different stages of a system. One fiber study combines

field data with offline sensing classification and illustrates how field collection and functional execution can occupy different parts of the evaluation path [133]. Segmented architectures add interfaces, timing, calibration, and conversion conditions to that path. When these elements connect both outcomes to one operating point, field coverage can support a stronger claim of joint operation.

C. Artifact Access and Reconstructability

Validation describes what a study tested. Reconstructability describes how completely another group can recover the operating point and the path from input to reported result. Access to data, code, or models supports this task, while physical experiments also depend on the system configuration, measurement conditions, processing steps, and provenance.

Table VI summarizes the access routes reported for data and for code or models in all 206 studies.

The reported routes support different forms of reconstruction. Open data appear across simulation, laboratory, prototype, computer vision, and field data settings [104], [123], [133], [202], [235]. They support reanalysis of the released data. Complete system reconstruction also requires configuration and execution details. Software preserves the transformation from inputs to reported outputs, and one computer vision study reported partial code or model components [104].

Reconstructing a physical experiment requires the system and hardware, the scene and calibration, the waveform and

TABLE VI
REPORTED ACCESS TO DATA AND TO CODE OR MODELS IN THE 206 STUDIES

Reported access	Studies, n (%)
<i>Data</i>	
Closed or status unreported	145 (70.4%)
Access by request	41 (19.9%)
Open access	13 (6.3%)
Outside artifact scope	7 (3.4%)
<i>Code or model</i>	
Closed or status unreported	197 (95.6%)
Access by request	7 (3.4%)
Partial components	1 (0.5%)
Outside artifact scope	1 (0.5%)

Note. The first row in each group combines closed access and an unspecified access status.

processing, the metrics and data, and the versioned release. These requirements extend the comparison profile in Section II. Electronic Supplements S-Data Dictionary and S-Evidence provide the field definitions and record provenance used in the review.

The TQAF reproducibility assessment reads artifact access together with parameter reporting. It places 4 studies in the low category, 199 in the adequate category, and 3 in the strong category. Electronic Supplement S-Appraisal provides scores for each study. Thirteen studies report open data, while three reach strong reproducibility. The difference reflects the additional reconstruction information captured by the appraisal.

D. Benchmark Readiness

A reconstructable result becomes a reusable benchmark when another group can apply the same task, condition set, baseline, and metric definition. The TQAF benchmark assessment combines these requirements with artifact access, validation, reproducibility, and comparison admissibility. The observed distribution comprises 48 studies in the low category, 158 in the adequate category, and zero in the strong category.

Under the review criteria, strong readiness depends on bringing an external or common baseline, an open artifact, admissible outcomes in both functional domains, and strong validation and reproducibility support into the same evidence chain. S-Appraisal reports these scores. This distribution directs attention toward shared tasks, baselines, artifacts, and evaluation conditions that other groups can reuse.

These distributions characterize evidence reach, functional coverage, artifact access, and reconstructability within the 206 included studies. They provide the descriptive basis for the benchmark and reporting actions in Section VIII.

Overall, the corpus combines a broad laboratory and prototype base with a smaller set of field and deployment results. Stronger cumulative evidence emerges when both functions are tied to one traceable configuration and the evaluation path is preserved through reusable artifacts. Section VII builds on this validation map by examining the technologies and applications supported by the reported evidence and their connection to 6G network roles.

VII. ENABLING TECHNOLOGIES, APPLICATION REQUIREMENTS, AND 6G NETWORK EVIDENCE

Validation shows how far a reported system was tested. We now follow the technologies that form communication and sensing observables into application requirements and 6G network roles. Because a study may contribute to several technology and application labels, their counts describe coverage within the corpus.

Figure 8 follows this evidence chain from optical generation and transport through waveform formation, spatial control, inference, selected application domains, and 6G network positioning.

The chain shows that network interpretation depends on continuity across the physical observable, the joint design choice, the application condition, and the reported network role. The displayed categories guide the reader through that relationship, while the following subsections retain the platform conditions and study evidence.

A. Technologies That Create and Shape Observables

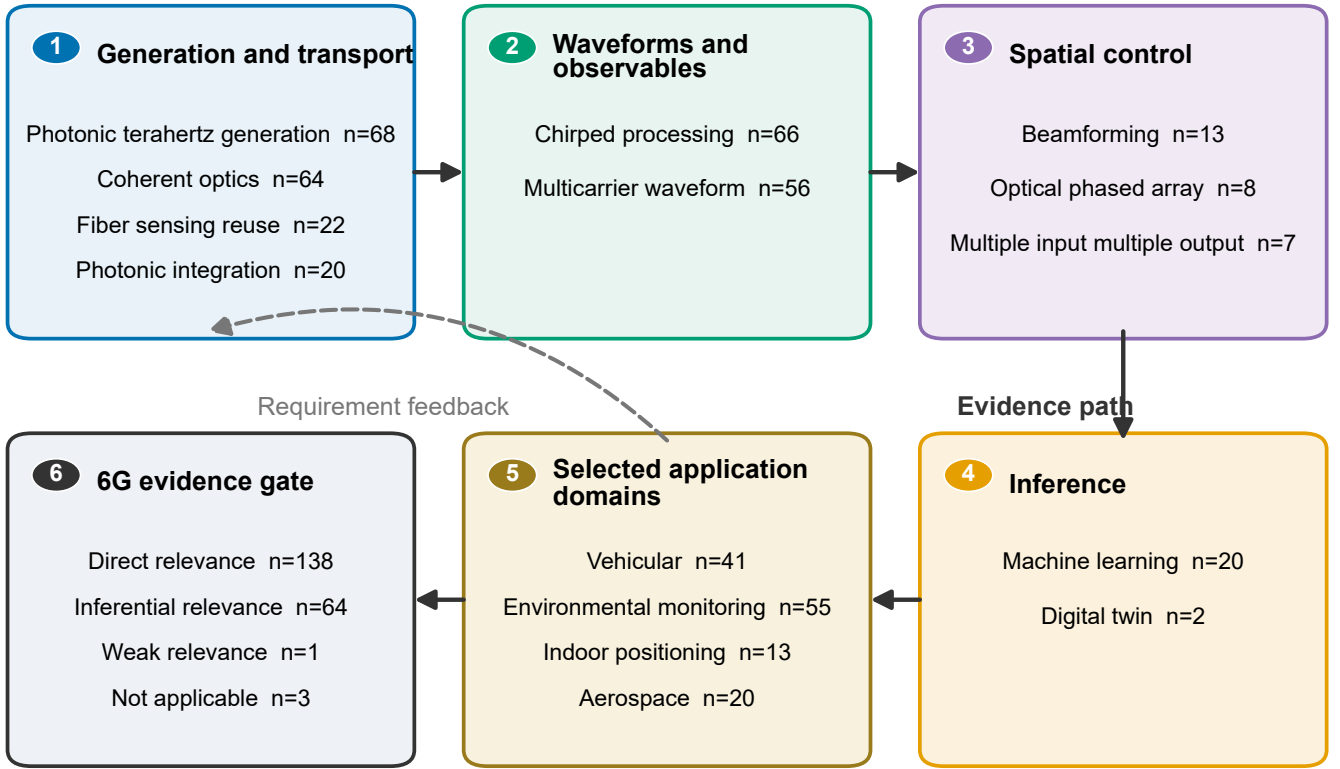
The main technology labels occupy complementary points in the signal chain. Photonic generation creates or translates the carrier. Chirped processing maps delay or motion to beat frequency. Coherent reception preserves phase, and OFDM provides controllable time and frequency resources. A single architecture can combine several of these mechanisms.

OFDM supports joint operation by placing payload and sensing resources within one signal structure. The sensing observable may come from pilots, payload tones, cyclic prefixes, reserved subcarriers, or a dedicated processing branch. This choice determines which samples are known at the receiver and how interference enters the estimate. Optical modulation constraints and the measurement plane used for SNR shape the interpretation. Embedded pilot fiber sensing and indoor optical positioning show how the same multicarrier principle serves different observables and geometries [98], [207].

Chirped and coherent systems draw on phase continuity and a shared timing reference. A returned chirp can reveal range or motion as the transmitted signal carries data. A coherent front end preserves small phase changes for distributed sensing and demodulation. Photonic generation can extend the reference to wideband millimeter wave or terahertz carriers. Full photonic dechirping and coherent fiber vibration sensing illustrate these two uses [56], [213]. Their performance depends on laser stability, reference distribution, conversion efficiency, receiver bandwidth, and calibration.

The optical stage governs carrier generation and reference stability. Once the carrier is radiated, the communication and sensing paths follow microwave, millimeter wave, or terahertz propagation. In a free space optical link, the optical beam itself traverses the channel. System evaluation should therefore follow the signal across both physical domains and identify where power, SNR, bandwidth, and timing are measured. This view makes the interface between components part of the joint system.

Spatial control changes both the communication link budget and the sensing geometry. Beamforming, optical phased



Technology and application labels may overlap. The 6G evidence gate assigns one class per study.

Fig. 8. Organization across system layers of the mechanisms that create and shape O-ISAC observables and connect them to service requirements. Technology and application counts describe overlapping corpus coverage, so a study may contribute to several displayed categories. The 6G evidence gate is exclusive and assigns each of the 206 studies to one relevance class. Selected application domains illustrate the requirement flow, while Table VII provides the complete application inventory.

arrays, multiple input multiple output processing, and reconfigurable optical intelligent surfaces provide different forms of spatial control. Beamforming directs energy. An optical phased array provides one steering mechanism. Multiple input multiple output processing uses several spatial dimensions, and an intelligent surface reshapes an intermediate propagation boundary. A system may combine these roles.

A narrow beam improves received power and angular selectivity while making acquisition, tracking, blockage, and calibration central to joint performance. The joint use of an optical phased array and OFDM and the use of frequency comb beamforming connect steering with ranging or imaging [82], [123]. Multiple input multiple output processing adds communication streams, sensing diversity, or separation among targets and paths. Structured fields and hybrid systems that combine free space optics with fiber Bragg gratings place these spatial dimensions at different points in the architecture [89], [94]. Meaningful comparison therefore requires the independent dimensions, geometry, and calibration method.

The intelligent surface evidence consists of one study that models a mobile free space optical path controlled by an optical intelligent surface [206]. A separate laboratory demonstration uses mechanically rotated Risley prisms to

steer a photonic generated terahertz signal [122]. Both mechanisms redirect energy, and each uses different surface physics, propagation laws, control variables, and validation settings. Their separate treatment preserves the physical meaning and provides two starting points for broader experimental comparison.

B. Integration Beyond the Waveform

Integration and inference move joint design beyond waveform selection. Photonic integration appears in 20 studies, fiber distributed acoustic sensing in 22, and machine learning or artificial intelligence in 20. Two studies use digital twins. A further 19 studies contain specialized mechanisms grouped as other because they fall outside the broad technology labels. Together, these categories cover device integration, infrastructure reuse, inference, and a diverse long tail.

Photonic integration can place generation, reference distribution, beam control, and detection on a compact hardware path. This arrangement reduces duplicated paths and brings packaging, thermal drift, crosstalk, and calibration into the same design problem. Microcomb synchronization coordinates wireless data with radar imaging. A distributed W band

system uses radar estimates for beam alignment and flexible data service [236], [237]. Both examples show how a shared reference can connect communication and sensing hardware.

Fiber distributed acoustic sensing follows a complementary route by using the transmission medium as a spatial sensor. Phase, backscatter, and polarization can reveal events along the route as the fiber carries traffic. Coherent multicore transmission with distributed sensing and operational submarine cable monitoring illustrate two scales of this infrastructure reuse [47], [161]. The central design question becomes how to preserve the optical observable together with traffic continuity.

Learning turns optical observations into event labels, positions, field estimates, channel predictions, and resource decisions. Each task is best read with its definition, reference data, training distribution, evaluation shift, uncertainty, and inference cost. Fiber diagnosis, federated visible light positioning, and wearable optical sensing show the range from distributed infrastructure to personal use [75], [133], [142].

A neural network can also take part in both functions. In one coherent link, the model compensates transmission impairments and estimates distributed power and abnormal loss [238]. The two outputs share the link while retaining separate targets and reference data. Digital twins add another model layer. One represents vehicular edge resources, and another provides a model aided physical reference for coherent fiber sensing [194], [234]. Clear reporting of the represented state, update process, mismatch model, and resulting decision gives these models an explicit system role.

C. Applications as Operating Requirements

Applications translate the mechanisms above into conditions that a joint service must satisfy. Table VII retains all thirteen application domains and maps each one to its joint role and defining evaluation conditions. A fiber platform for vehicles, for example, carries data at high rates and measures distributed temperature [239]. Devices used by people connect communication or remote alerts with motion, falls, air pressure, physiology, or gesture [75], [240], [241]. Each application therefore changes what the experiment must demonstrate, even when both functions share the same optical interface.

Across the map, the dominant requirement moves with the application. Network and mobility settings emphasize acquisition, timing, and continuity. Deployed infrastructure emphasizes persistent observation alongside carried traffic. Human use and extreme settings add physiological references, safety, privacy, pointing, atmosphere, packaging, and access to the test environment. These requirement bundles show which operating conditions must accompany a reported result before it can support comparison or reuse.

D. From 6G Relevance to Network Evidence

The 6G relevance gate in Figure 8 assigns each study one corpus position. Direct cases connect the reported system explicitly to a 6G role, as illustrated by [55], [123]. Inferential cases contribute an enabling link that gains meaning from additional network context, as illustrated by [56], [98]. The remaining classes capture a lighter connection to 6G or an

application focus outside that framing, as illustrated by [181], [241].

The distribution connects the reviewed corpus with 6G access, transport, localization, automation, environmental awareness, and generation of carriers at high frequencies. A relevance code therefore locates a study within the 6G discussion. Standard conformance, interoperability, and deployment readiness require additional evidence tied to the relevant network framework [4].

A stronger network interpretation emerges when the optical or photonic subsystem occupies an explicit role across the complete network path. Access, fronthaul, transport, edge processing, and service control bring timing, mobility, orchestration, energy, security, and recovery requirements. Evidence at this level evaluates useful communication and sensing under the conditions of that role and reports the resources needed to sustain both. This creates a traceable path from the physical observable and hardware to the service objective.

Within this survey, 6G relevance is strongest when both functional outcomes and their operating conditions remain traceable to a network role. The resulting evidence chain links physical observables, application requirements, and service objectives and provides the basis for the research priorities in Section VIII.

VIII. DISCUSSION, RESEARCH ROADMAP, AND LIMITATIONS

Across the evidence base, the central issue is whether physical sharing produces joint outcomes that remain traceable to one operating context. This section interprets that relationship, develops five research priorities, and states the review boundaries.

A. Interpretation Across the Evidence Base

Physical integration identifies where communication and sensing share infrastructure, signals, or information. Evidential integration shows how clearly both outcomes remain tied to the same operating conditions. Together, they connect architectural and transmission perspectives while preserving the physical regime of each platform [8], [9], [19], [22].

Deeper sharing can connect more design variables and improve coordination across the two functions. It can also couple power, distortion, synchronization, dynamic range, and failure behavior. Its scientific value therefore depends on locating these effects in the signal chain and tracing their influence on both outcomes.

Tradeoff patterns transfer across studies when tasks, metric definitions, measurement planes, baselines, constraints, and validation settings remain aligned. This alignment allows model bounds and empirical operating points to retain their distinct roles and support comparison within a stated context [12], [13].

Fielded cable and network studies illustrate the value of operational continuity [16], [18]. Within the survey, system evidence becomes reusable when configuration, calibration, processing, and reported artifacts remain connected across interfaces. This continuity links physical sharing to application

TABLE VII
APPLICATION DOMAINS AND OPERATING REQUIREMENTS IN THE REVIEWED CORPUS

<i>Network and mobility</i>			
Domain, n (%)	Common communication and sensing role	Key operating condition	Studies
6G and access 100 (48.5%)	Access and transport with channel, target, position, or environment sensing	Timing, orchestration, continuity, and evidence for both functions	[55], [89]
Vehicular 41 (19.9%)	Vehicle data links with range, motion, pose, temperature, or channel sensing	Changing geometry, acquisition, tracking, and service continuity	[63], [239]
Optical access 32 (15.5%)	Access and transport links with channel, event, or fiber state sensing	Reach, synchronization, and traffic continuity	[55], [98], [133]
Indoor positioning 13 (6.3%)	Indoor optical links with user or robot position and orientation sensing	Geometry, calibration, coverage, and update continuity	[61], [207]
Data center 4 (1.9%)	Short reach data links with channel estimation or link state sensing	Traffic continuity and calibrated link references	[98]
<i>Infrastructure</i>			
Domain, n (%)	Common communication and sensing role	Key operating condition	Studies
Environmental monitoring 55 (26.7%)	Distributed links with vibration, temperature, or environmental event sensing	Persistent observation, localization, and traceable reference data	[47], [133], [221]
Smart infrastructure 41 (19.9%)	Infrastructure links with structural, motion, or event monitoring	Continuous observation, calibration, and resilient alert delivery	[89], [133], [139]
Industrial 34 (16.5%)	Industrial data or control with process, strain, temperature, motion, or position sensing	Delay, continuity, calibration, and safe operation	[75], [89]
Security and surveillance 26 (12.6%)	Monitoring links with anomaly, event, or location sensing	Coverage, event discrimination, and alert latency	[133]
<i>Human use, extreme settings, and remaining domains</i>			
Domain, n (%)	Common communication and sensing role	Key operating condition	Studies
Healthcare 6 (2.9%)	Wearable or ambient links with vital sign, activity, or gesture sensing	Motion robustness, physiological reference, illumination, and user safety	[74], [75]
Aerospace 20 (9.7%)	Satellite links with atmospheric or wavefront sensing	Pointing, turbulence, delay, and link adaptation	[87]
Underwater 2 (1.0%)	Underwater link concepts and subsea data links with temperature or event sensing	Water attenuation or cable calibration, packaging, and access	[139], [221]
Other 17 (8.3%)	Emerging links with range, imaging, object, or channel sensing	Metrics and reporting specific to the task	[122]

Note. Application domains overlap, so a study may contribute to more than one row. Counts and percentages describe coverage within the 206 included studies. For each domain, the table includes a selected study set and lists the corresponding references in the final column. The 6G and access row identifies the application category, and the final subsection provides the 6G relevance classification.

and network claims and motivates a central research task. Joint operating points should remain interpretable, testable, and reusable across groups.

B. Research Priorities for Reusable Evidence

Table VIII organizes five priorities and pairs each observed finding with an action and a measurable sign of progress.

The order matters because each stage uses evidence produced by the previous one. Clear measurement context supports fair benchmarks. Benchmarks support realistic joint tests. Reconstructable artifacts make those tests repeatable, and repeatable tests provide a basis for adaptive and secure evaluation at network scale.

Stress tests should follow the native signal path of each modality. Fiber tests can vary topology and traffic, free space optical tests can vary geometry and atmosphere, and visible light tests can vary illumination and orientation. Photonic terahertz tests can vary optical drift together with wireless propagation. Matched baselines for the individual functions keep the limiting mechanism visible across these settings.

For standards and 6G coordination, the same sequence can support a core reporting profile that preserves modality

physics, ground truth, and reusable benchmark definitions. Shared metadata can connect results across platforms while retaining the conditions needed for interpretation.

C. Scope and Limitations of the Review

The review represents English language technical reports published from 1 January 2020 through 22 June 2026 and indexed by the six searched sources. Full texts were available and assessed for 272 of the 330 reports sought, which corresponds to 82.4% retrieval coverage. Two Taylor & Francis exports with few records are documented through their complete platform exports. These boundaries define the literature represented by the final corpus of 206 studies.

Searches and screening preceded retrospective registration, and the subsequent search refresh and report reconciliation produced the final corpus described in Section III. One investigator supervised routine decisions, while independent checks focused on specified evidence records and report reconciliation.

TQAF provides a technical appraisal developed for this review. The diversity of tasks, quantities, validation settings, outcomes, and source protocols supports structured narrative

TABLE VIII
RESEARCH PRIORITIES DERIVED FROM THE REVIEWED EVIDENCE

Priority	Finding and proposed action	Evidence of progress
1. Measurement contract	<i>Finding.</i> Of 4,779 metric records, 118 supported bounded relations under aligned conditions. <i>Action.</i> Record the task, plane, unit, baseline, operating condition, validation setting, and source locator.	An independent reader reconstructs the quantity and condition from a structured record.
2. Benchmark for each modality	<i>Finding.</i> Comparison across studies depends on matched physical context. <i>Action.</i> Repeat one task with reference and independent designs under shared calibration and ground truth.	Independent groups reproduce the relative finding under the shared protocol.
3. Joint operation under disturbance	<i>Finding.</i> Of 402 substantive tradeoff relationships, 371 depend on reported conditions. Six of 12 field or deployment tier studies report outcomes in both domains. Traceability is strongest when both outcomes share the same timing and condition set. <i>Action.</i> Evaluate both functions in the same run and condition set under load, a disturbance relevant to the modality, and matched baselines for the individual functions.	Repeated trials recover the joint region and its limiting mechanism.
4. Reconstructable experiment package	<i>Finding.</i> Open data appear in 13 of 206 studies, and code or model access reaches a partial release in one. <i>Action.</i> Preserve a versioned package and rerun the main comparison under one added condition.	An independent rerun recovers the comparison and added condition from the package.
5. Intelligence, scale, and security	<i>Finding.</i> Technology and application coverage connects the corpus to 6G roles. Conformance, interoperability, and readiness become measurable through explicit network tests. <i>Action.</i> Apply a declared shift, attack, load, or component failure and report overhead, resource use, decisions, and fallback.	The service retains an explainable boundary with resource use, decisions, and fallback recorded.

Note. The findings summarize the retained corpus presented in Sections IV through VII. The proposed actions retain the physical conditions of each modality and define observable signs of progress.

synthesis, which preserves native conditions and bounded comparisons. This analytical scale centers interpretation on reported mechanisms and conditions. Harmonized outcomes and source protocols form the appropriate basis for pooled effects, sensitivity estimates, and causal moderator analysis.

The multilabel taxonomy incorporates reviewer judgment, especially for hybrid systems. Row counts represent coded observations at extraction granularity, while effect analysis uses a different unit. For 92 metric rows, the available fields supported interpretation within the source setting and a conservative comparison status. Artifact access combined source reporting with targeted link checks, so the appraisal reflects documented reconstruction information.

The conclusions concern recurring mechanisms, measurement constraints, and validation patterns within the retained corpus. Corpus distributions describe the reviewed literature. Worldwide prevalence, market maturity, and platform rank

require evidence designed for those questions. Within its stated scope, the review identifies the conditions that allow optical integration to support transferable joint results.

IX. CONCLUSION

Across the reviewed corpus, O-ISAC is organized most clearly by the physical role of optics and the mechanism that couples communication and sensing. Optics may carry the user link, illuminate the target, distribute a carrier, or preserve a coherent reference, and each role creates a different set of constraints. Shared hardware, carriers, waveforms, resources, links, and design rules then connect the two functions at different points in the system. Their engineering value depends on how the task, metric definition, measurement plane, operating condition, and validation setting are reported together.

The evidence indicates that recurring tradeoff mechanisms can be compared when their physical context and measurement contracts align. Laboratory and prototype studies provide a broad experimental base, while field results illustrate how topology, geometry, propagation, calibration, and service continuity enter real systems. Application and 6G relevance become more informative when both functional outcomes remain traceable to the same network role and condition set. This makes transferable joint evidence the central measure of progress and keeps individual performance values tied to their source conditions.

Progress therefore depends on measurement contracts tailored to each modality, paired validation under realistic disturbances, and versioned artifacts that preserve configuration, calibration, data, and processing. These practices provide a path from individual demonstrations to reusable benchmarks and evaluation at network scale. They also allow subsequent designs to build on prior evidence while preserving the physical meaning of each optical platform.

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